

# Safety Data Sheet

according to Regulation (EC) No 1907/2006



Trade name: LPKF ProConduct Paste  
Revision date: 19.06.2015 Version: 3.0  
Print date: 19.06.2015 Replaces version: 2.1

## SECTION 1: Identification of the substance/mixture and of the company

### 1.1 Product identifier

LPKF ProConduct Paste

### 1.2 Relevant identified uses of the substance or mixture

Through-hole plating of PCBs

### 1.3 Details of the supplier of the safety data sheet

#### Supplier

LPKF Laser & Electronics AG

#### Street/P.O. Box

Osteriede 7

#### Country Identifier/Postal Code/City

D-30827 Garbsen

#### Information contact

Product management

#### Telephone/Fax/Email

+49 5131 7095-0 / +49 5131 7095-90 / info@lpkf.com

### 1.4 Emergency telephone number

Telephone number (24/7): +49 551 38318-66

## SECTION 2: Hazards identification

### 2.1 Classification of the substance or mixture

#### Classification according to Regulation (EC) No. 1272/2008:

|                                            |            |                                                            |
|--------------------------------------------|------------|------------------------------------------------------------|
| Skin irritation                            | Category 2 | H315 Causes skin irritation.                               |
| Serious eye irritation                     | Category 2 | H319 Causes serious eye irritation.                        |
| Germ cell mutagenicity                     | Category 2 | H341 Suspected of causing genetic defects                  |
| Acute hazards to the aquatic environment   | Category 1 | H400 Very toxic to aquatic life.                           |
| Chronic hazards to the aquatic environment | Category 1 | H410 Very toxic to aquatic life with long lasting effects. |

#### Classification according to Directive 67/548/EEC or 1999/45/EC:

Mutagenic Cat. 3  
R68 Possible risk of irreversible effects.  
Xi - Irritant  
R36/38 Irritating to eyes and skin.  
N - Dangerous for the environment  
R50/53 Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

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## 2.2 Label elements

### Label elements

according to Regulation (EC) No. 1272/2008

Pictogram:



Signal word / Indication of danger:

Warning

### Hazard Statements

H315 Causes skin irritation.  
H319 Causes serious eye irritation.  
H341 Suspected of causing genetic defects.  
H410 Very toxic to aquatic life with long-lasting effects.

### Precautionary Statements

P280 Wear protective gloves/eye protection.

### Other label elements

## 2.3 Other hazards

None if used for intended purpose.

## SECTION 3: Composition/information on ingredients

### 3.1 Mixtures

#### Base substances of the mixture:

Pigment  
Solvent

**Substance name:** Silver

EC No.: 231-131-3 CAS No.: 7440-22-4 Index No.:

**% by weight:** 60- 80 %

Classification according to 67/548/EEC:

Classification according to Regulation (EC) No. 1272/2008:

REACH registration number:

N - Dangerous for the environment;  
R50/53

Aquatic Acute 1, H400

Aquatic Chronic 1, H410

M factor: 1.000 M factor (Chron

Aquat Tox): 1.000

**Substance name:** 2-(2-Ethoxyethoxy)ethyl acetate

EC No.: 203-940-1 CAS No.: 112-15-2 Index No.:

**% by weight:** 10 – 20%

Classification according to 67/548/EEC:

Classification according to Regulation (EC) No. 1272/2008:

REACH Registration No.:

Xi - Irritant; R36

Eye Irrit. 2, H319

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**Substance name:** Phenol  
EC No.: 203-632-7 CAS No.: 108-95-2 Index No.:  
REACH registration number: 01-2119471329-32  
**% by weight:** 1 - < 3 %  
Classification according to 67/548/EEC:

Mutagen category 3.; R68  
T - Toxic; R23/24/25  
C - Corrosive; R34  
Xn - Harmful; R48/20/21/22  
Muta. 2, H341  
STOT RE 2, H373  
Skin Corr. 1B, H314  
Acute Tox. 3; Dermal, H311  
Acute Tox. 3; Oral, H301  
Acute Tox. 3; Inhalation, H331

Classification according to Regulation (EC) No. 1272/2008:

**Substance name:** Solvent Naphtha (petroleum), light arom., <0.1% Benzene  
EC No.: 265-199-0 CAS No.: 64742-95-6 Index No.:  
REACH registration number: 01-2119455851-35  
**% by weight:** 0.1 - 1 %  
Classification according to 67/548/EEC:

N - Dangerous for the environment;  
R51/53  
Xn - Harmful; R65  
Xi - Irritant; R37  
R10, R66, R67  
n.a.

Classification according to Regulation (EC) No. 1272/2008:

**Substance name:** 4-(1,1,3,3-tetramethylbutyl)phenol  
EC No.: 205-426-2 CAS No.: 140-66-9 Index No.:  
REACH registration number: 01-2119541687-29  
**% by weight:** < 0.1 %  
Classification according to 67/548/EEC:

N - Dangerous for the environment; R50, R53  
Xi - Irritant; R38, R41  
Skin Irrit. 2, H315  
Eye Dam. 1, H318  
Aquatic Acute 1, H400  
Aquatic Chronic 1, H410  
=====  
EU. REACH Candidate List of Substances of Very High Concern for Authorization (SVHC)  
M factor: 10 M factor (Chron Aquat Tox): 10

Classification according to Regulation (EC) No. 1272/2008:

The product contains a substance of very high concern below the limit value (< 0.1 %) relevant for classification.  
Substances without classification may have community workplace exposure limits available.  
(See section 16 for the full text of the hazard statements and R phrases.)

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## **SECTION 4: First aid measures**

### **4.1 Description of first aid measures**

**In case of inhalation**

Fresh air, consult doctor.

**In case of skin contact**

IF ON SKIN: Wash with plenty of soap and water.

**Following eye contact**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

**Following ingestion**

Rinse mouth, drink 1-2 glasses of water, do not induce vomiting, consult a doctor.

### **4.2 Most important symptoms and effects, both acute and delayed**

Skin: Redness, inflammation.

Eye: Irritation, conjunctivitis

Prolonged or repeated skin contact with silver and its salts may cause a blue-grey discoloration of the skin and mucous membranes that is irreversible (Argyria).

### **4.3 Indication of any immediate medical attention and special treatment needed**

See section: Description of first aid measures

## **SECTION 5: Firefighting measures**

### **5.1 Extinguishing media**

**Suitable:**

Carbon dioxide, foam, powder

**Unsuitable:**

High pressure water jet

### **5.2 Special hazards arising from the substance or mixture**

Formation of toxic gases is possible during heating or in fires.

### **5.3 Advice for firefighters**

Wear protective equipment.

### **5.4 Additional information:**

In case of fire, keep containers cool with water spray.

## **SECTION 6: Accidental release measures**

### **6.1 Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation.

Avoid contact with skin and eyes.

### **6.2 Environmental precautions**

Do not empty into drains / surface water / ground water.

### **6.3 Methods and material for containment and cleaning up**

Remove with liquid-absorbing material (sand, peat, sawdust).

Dispose of contaminated material as waste according to Section 13.

### **6.4 Reference to other sections**

Observe advice in section 8.

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## SECTION 7: Handling and Storage

### 7.1 Precautions for safe handling

Avoid skin and eye contact.  
Ensure that workrooms are adequately ventilated.  
Observe advice in section 8.  
**Hygiene measures:**  
Wash hands before work breaks and after finishing work.  
Do not eat, drink or smoke while working.  
Take off contaminated clothing and wash before reuse.

### 7.2 Conditions for safe storage, including any incompatibilities

Ensure good ventilation/extraction.  
For optimum shelf life store in original containers under refrigerated conditions at 4 °C to 8 °C.  
Store in a cool, well-ventilated place.  
**Storage class:** 6.1C

### 7.3 Specific end uses

See section 1.2

## SECTION 8: Exposure controls / personal protection

### 8.1 Control parameters

#### 8.1.1 Occupational limits and/or biological limit values

##### Occupational exposure limits (AGW) Germany

| Ingredient<br>CAS No.                      | ppm | mg/m <sup>3</sup> | Type                                     | Category                                                | Remarks  |
|--------------------------------------------|-----|-------------------|------------------------------------------|---------------------------------------------------------|----------|
| Silver, metallic<br>7440-22-4              |     | 0.1               | Time Weighted<br>Average (TWA)           | Indicative                                              | ECTLV    |
| Silver, inhalable<br>fraction<br>7440-22-4 |     |                   | Short Term<br>Exposure<br>Classification | Category II:<br>substances with a<br>resorptive effect. | TRGS 900 |
| Silver, inhalable<br>fraction<br>7440-22-4 |     | 0.1               | Exposure limit(s)                        | 8                                                       | TRGS 900 |
| Phenol<br>108-95-2                         | 2   | 8                 | Time Weighted<br>Average (TWA)           | Indicative                                              | ECTLV    |
| Phenol<br>108-95-2                         | 4   | 16                | Short Term<br>Exposure Limit<br>(STEL):  | Indicative                                              | ECTLV    |
| Phenol<br>108-95-2                         |     |                   | Skin designation                         | Can be absorbed<br>through the skin.                    | TRGS 900 |
| Phenol<br>108-95-2                         | 2   | 8                 | Exposure limit(s)                        | 2                                                       | TRGS 900 |
| Phenol<br>108-95-2                         |     |                   | Short Term<br>Exposure<br>Classification | Category II:<br>substances with a<br>resorptive effect. | TRGS 900 |

##### Predicted No-Effect Concentration (PNEC):

| Name on list       | Environmental<br>Compartment | Exposure<br>period | Value       | Remarks |
|--------------------|------------------------------|--------------------|-------------|---------|
| Phenol<br>108-95-2 | aqua (freshwater)            |                    | 0.077 mg/L  |         |
| Phenol<br>108-95-2 | aqua (marine water)          |                    | 0.0077 mg/L |         |

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| Name on list                                                 | Environmental<br>Compartment    | Exposure<br>period | Value         | Remarks |
|--------------------------------------------------------------|---------------------------------|--------------------|---------------|---------|
| Phenol<br>108-95-2                                           | sediment<br>(freshwater)        |                    | 0.0915 mg/kg  |         |
| Phenol<br>108-95-2                                           | sediment (marine<br>water)      |                    | 0.00915 mg/kg |         |
| Phenol<br>108-95-2                                           | soil                            |                    | 0.136 mg/kg   |         |
| Solvent naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | aqua (freshwater)               |                    | 0.635 mg/L    |         |
| Solvent naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | aqua (marine water)             |                    | 0.0635 mg/L   |         |
| Solvent naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | aqua (intermittent<br>releases) |                    | 6.35 mg/L     |         |
| Solvent naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | STP                             |                    | 100 mg/L      |         |
| Solvent naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | sediment<br>(freshwater)        |                    | 3.29 mg/kg    |         |
| Solvent naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | sediment (marine<br>water)      |                    | 0.329 mg/kg   |         |
| Solvent naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | soil                            |                    | 0.29 mg/kg    |         |

## Derived No-Effect Level (DNEL):

| Name from list                                               | Scope   | Exposure<br>route | Health<br>effects                              | Exposure<br>time | Value                 |
|--------------------------------------------------------------|---------|-------------------|------------------------------------------------|------------------|-----------------------|
| Phenol<br>108-95-2                                           | workers | dermal            | Long term<br>exposure -<br>systemic<br>effects |                  | 1,23 mg/kg<br>bw/day  |
| Phenol<br>108-95-2                                           | workers | Inhalation        | Long term<br>exposure -<br>systemic<br>effects |                  | 8 mg/m <sup>3</sup>   |
| Solvent Naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | workers | dermal            | Long-term<br>exposure -<br>systemic<br>effects |                  | 25 mg/kg<br>bw/day    |
| Solvent Naphtha<br>(petroleum), light<br>arom.<br>64742-95-6 | workers | Inhalation        | Long-term<br>exposure -<br>systemic<br>effects |                  | 150 mg/m <sup>3</sup> |

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| Name from list                                      | Scope              | Exposure route | Health effects                        | Exposure time | Value                |
|-----------------------------------------------------|--------------------|----------------|---------------------------------------|---------------|----------------------|
| Solvent Naphtha (petroleum), light arom. 64742-95-6 | general population | Inhalation     | Long-term exposure - systemic effects |               | 32 mg/m <sup>3</sup> |
| Solvent Naphtha (petroleum), light arom. 64742-95-6 | general population | dermal         | Long-term exposure - systemic effects |               | 11 mg/kg bw/day      |
| Solvent Naphtha (petroleum), light arom. 64742-95-6 | general population | oral           | Long-term exposure - systemic effects |               | 11 mg/kg bw/day      |

## Biological limit values:

| Ingredient      | Parameter              | Test material       | Test date:                   | Conc.    | Basis of the limit value |
|-----------------|------------------------|---------------------|------------------------------|----------|--------------------------|
| PHENOL 108-95-2 | Phenol                 | Urine               | Sampling time: End of shift. | 300 mg/l | DE BAT                   |
| PHENOL 108-95-2 | Phenol with hydrolysis | Creatinine in urine | Sampling time: End of shift. | 120 mg/g | DE BAT                   |

## 8.2 Exposure controls

### 8.2.1 Appropriate engineering controls

Ensure good ventilation/extraction at the workplace.

### 8.2.2 Individual protection measures - personal protection equipment

#### Eye/face protection

Protective goggles

#### Skin protection

##### Gloves

##### Full contact:

##### Glove material:

Chemical-resistant protective gloves (EN 374).  
Suitable materials (recommended: at least protection index 6): Isobutylene-isoprene rubber

##### Glove thickness (mm):

>= 0.7 mm

##### Break through time (min.)

> 480

##### Splash contact:

##### Glove material:

Chemical-resistant protective gloves (EN 374).  
Suitable materials (recommended: at least protection index 2): Isobutylene-isoprene rubber

##### Glove thickness (mm):

>= 0.7 mm

##### Break through time (min.)

> 30

This information is based on literature references and on information provided by glove manufacturers, or is derived by analogy with similar substances. Please note that in practice the working life of chemical-resistant protective gloves may be considerably shorter than the permeation time determined in accordance with EN 374 as a result of the many influencing factors (e.g. temperature). If signs of wear and tear are noticed then the gloves should be replaced.

#### Body protection

Wear suitable protective clothing.

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## Respiratory protection

In case of aerosol formation, we recommend wearing of appropriate respiratory protection equipment with ABEK P2 filter.

This recommendation has to be adapted to local conditions.

## SECTION 9: Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

|                                               |                 |                                    |
|-----------------------------------------------|-----------------|------------------------------------|
| Appearance                                    | Physical state: | Liquid                             |
|                                               |                 | Liquid                             |
|                                               | Colour:         | silver                             |
| Odour:                                        |                 | Solvent                            |
| pH value:                                     |                 | Not applicable                     |
| Melting/freezing point:                       |                 | No data available / not applicable |
| Initial boiling point and range:              |                 | 217 °C (422.6 °F)                  |
| Flash point                                   |                 | 110 °C (230 °F); no method         |
| Evaporation rate:                             |                 | No data available / not applicable |
| Flammability (solid, gas):                    |                 | No data available / not applicable |
| Upper/lower flammability or explosive limits: |                 | 1.0 %(V)<br>19.4 %(V)              |
| Vapour pressure:                              |                 | No data available / not applicable |
| Relative density:                             |                 | 2.68 g/cm <sup>3</sup>             |
| Bulk density:                                 |                 | No data available / not applicable |
| Solubility(ies):                              |                 | Insoluble in water                 |
| Partition coefficient n-octanol/water:        |                 | No data available / not applicable |
| Auto-ignition temperature:                    |                 | No data available / not applicable |
| Decomposition temperature:                    |                 |                                    |
| Viscosity:                                    |                 | No data available / not applicable |
| Explosive properties:                         |                 | No data available / not applicable |
| Oxidising properties:                         |                 | No data available / not applicable |

### 9.2 Other information

No data available / not applicable

## SECTION 10: Stability and Reactivity

### 10.1 Reactivity

Reaction with strong oxidants.

### 10.2 Chemical stability

Stable under recommended storage conditions.

### 10.3 Possibility of hazardous reactions

See section reactivity

### 10.4 Conditions to avoid

No decomposition if used according to specifications.

### 10.5 Incompatible materials

See section reactivity

### 10.6 Hazardous decomposition products

None known if used for intended purpose.  
In case of fire toxic gases can be released.



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## SECTION 11: Toxicological information

### 11.1 Information on toxicological effects

#### General toxicological information

The mixture is classified based on the available hazard information for the ingredients as defined in the classification criteria for mixtures for each hazard class or differentiation in Annex I to Regulation 1272/2008/EC. Relevant available health/ecological information for the substances listed under Section 3 is provided in the following.

#### Skin irritation

Causes skin irritation.

#### Eye irritation

Causes serious eye irritation.

#### Mutagenicity

Suspected of causing genetic defects

#### Acute toxicity

| Hazardous ingredients CAS-No.                       | Value type | Value         | Route of exposure    | Exposure time | Species | Method                                         |
|-----------------------------------------------------|------------|---------------|----------------------|---------------|---------|------------------------------------------------|
| Silver >= 99,9 % Ag in powder (< 1 mm)<br>7440-22-4 | LD50       | > 2,000 mg/kg | oral                 |               | Rat     | OECD Guideline 401 (Acute Oral Toxicity)       |
| Phenol<br>108-95-2                                  | LC0        |               | inhalative (Aerosol) | 8 h           | Rat     | OECD Guideline 403 (Acute Inhalation Toxicity) |

#### Skin corrosion/Irritation

| Hazardous ingredients CAS-No.               | Result              | Exposure duration | Species | Method                                                   |
|---------------------------------------------|---------------------|-------------------|---------|----------------------------------------------------------|
| 2-(2-Ethoxyethoxy)ethyl acetate<br>112-15-2 | slightly irritating |                   | Rabbit  | OECD Guideline 404 (Acute Dermal Irritation / Corrosion) |
| Phenol<br>108-95-2                          | corrosive           | 3 min             |         |                                                          |

#### Serious eye damage/eye irritation

| Hazardous ingredients CAS-No.               | Result     | Exposure duration | Species | Method                                                |
|---------------------------------------------|------------|-------------------|---------|-------------------------------------------------------|
| 2-(2-Ethoxyethoxy)ethyl acetate<br>112-15-2 | irritating |                   | Rabbit  | OECD Guideline 405 (Acute Eye Irritation / Corrosion) |
| Phenol<br>108-95-2                          | corrosive  |                   | Rabbit  | OECD Guideline 405 (Acute Eye Irritation / Corrosion) |

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## Germ cell mutagenicity

| Hazardous ingredients CAS-No. | Result   | Study type / route of administration      | Metabolic activation/ exposure time | Species | Method |
|-------------------------------|----------|-------------------------------------------|-------------------------------------|---------|--------|
| Phenol 108-95-2               | positive | in vitro mammalian cell micronucleus test | with and without                    |         |        |

## Carcinogenicity

| Hazardous ingredients CAS-No. | Result           | Species | Sex         | Exposure time / Frequency of treatment | Route of application | Method                                       |
|-------------------------------|------------------|---------|-------------|----------------------------------------|----------------------|----------------------------------------------|
| Phenol 108-95-2               | not carcinogenic | mouse   | male/female | 103 w daily ad libitum (continuous)    | oral: drinking water | OECD Guideline 451 (Carcinogenicity Studies) |

## Reproductive toxicity:

| Hazardous ingredients CAS-No. | Result / Classification                                            | Study type / route of administration         | Exposure time | Species | Method                                                          |
|-------------------------------|--------------------------------------------------------------------|----------------------------------------------|---------------|---------|-----------------------------------------------------------------|
| Phenol 108-95-2               | NOAEL P = 71 mg/kg<br>NOAEL F1 = 70 mg/kg<br>NOAEL F2 = 1.000 mg/l | two-generation study<br>oral: drinking water |               | rat     | OECD Guideline 416 (Two-Generation Reproduction Toxicity Study) |

## SECTION 12: Ecological information

### General ecological information

The mixture is classified based on the available hazard information for the ingredients as defined in the classification criteria for mixtures for each hazard class or differentiation in Annex I to Regulation 1272/2008/EC. Relevant available health/ecological information for the substances listed under Section 3 is provided in the following.

### Other adverse effects

Do not empty into drains, soil or bodies of water.

## 12.1 Toxicity

### Ecotoxicity

Very toxic to aquatic life.  
Very toxic to aquatic life with long lasting effects.

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| Hazardous ingredients<br>CAS-No.                    | Value<br>type | Value          | Study of<br>acute<br>Toxicity | Exposure<br>time | Species                                                                             | Method                                                                          |
|-----------------------------------------------------|---------------|----------------|-------------------------------|------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Phenol<br>108-95-2                                  | LC50          | 21.93<br>mg/l  | Fish                          | 14 d             | Poecilia<br>reticulata                                                              | OECD<br>Guideline 204<br>(Fish,<br>Prolonged<br>Toxicity Test:<br>14-day Study) |
| Phenol<br>108-95-2                                  | LC50          | 24.9 mg/l      | Fish                          | 96 h             | Pimephales<br>promelas                                                              |                                                                                 |
| Phenol<br>108-95-2                                  | EC50          | 3.1 mg/l       | Daphnia                       | 48 h             | Ceriodaphnia<br>dubia                                                               |                                                                                 |
| Phenol<br>108-95-2                                  | EC50          | 61.1 mg/l      | Algae                         | 96 h             | Pseudokirchner<br>ella subcapitata<br>(reported as<br>Selenastrum<br>capricornutum) |                                                                                 |
| 4-(1,1,3,3-Tetra-<br>methylbutyl)phenol<br>140-66-9 | LC50          | 0.26 mg/l      | Fish                          | 96 h             | Leuciscus idus<br>melanotus                                                         | EU Method<br>C.1 (Acute<br>Toxicity for<br>Fish)                                |
| 4-(1,1,3,3-Tetra-<br>methylbutyl)phenol<br>140-66-9 | NOEC          | 0.0061<br>mg/l | Fish                          | 60 d             | Salmo gairdneri<br>(new name:<br>Oncorhynchus<br>mykiss)                            | OECD 210<br>(fish early lite<br>stage toxicity<br>test)                         |
| 4-(1,1,3,3-Tetra-<br>methylbutyl)phenol<br>140-66-9 | LC50          | 0.12 mg/l      | Fish                          | 14 d             | Salmo gairdneri<br>(new name:<br>Oncorhynchus<br>mykiss)                            | OECD<br>Guideline 204<br>(Fish,<br>Prolonged<br>Toxicity Test:<br>14-day Study) |
| 4-(1,1,3,3-Tetra-<br>methylbutyl)phenol<br>140-66-9 | NOEC          | 0.11 mg/l      | Daphnia                       | 48 h             | Daphnia<br>magna                                                                    | OECD<br>Guideline 202<br>(Daphnia sp.<br>Acute<br>Immobilisation<br>Test)       |
| 4-(1,1,3,3-Tetra-<br>methylbutyl)phenol<br>140-66-9 | EC50          | 1.1 mg/l       | Algae                         | 72 h             | Scenedesmus<br>subspicatus<br>(new name:<br>Desmodesmus<br>subspicatus)             | OECD<br>Guideline 201<br>(Alga, Growth<br>Inhibition Test)                      |
| 4-(1,1,3,3-Tetra-<br>methylbutyl)phenol<br>140-66-9 | EC10          | 0.3 mg/l       | Algae                         | 72 h             | Scenedesmus<br>subspicatus<br>(new name:<br>Desmodesmus<br>subspicatus)             | OECD<br>Guideline 201<br>(Alga, Growth<br>Inhibition Test)                      |
| 4-(1,1,3,3-Tetra-<br>methylbutyl)phenol<br>140-66-9 | NOEC          | 0.03 mg/l      | chronic<br>Daphnia            | 21 d             | Daphnia<br>magna                                                                    | OECD 211<br>(Daphnia<br>magna,<br>Reproduction<br>Test)                         |

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## 12.2 Persistence and degradability

| Hazardous ingredients<br>CAS-No.               | Result                | Route of exposure | Degradability | Method                                                                |
|------------------------------------------------|-----------------------|-------------------|---------------|-----------------------------------------------------------------------|
| Phenol<br>108-95-2                             | readily biodegradable | aerobic           | 62 %          | OECD Guideline 301 C (Ready Biodegradability: Modified MITI Test (I)) |
| 4-(1,1,3,3-tetramethylbutyl)phenol<br>140-66-9 |                       | aerobic.          | 0 %           | ISO 10708 (BODIS-Test)                                                |

## 12.3 Bioaccumulative potential / mobility in soil

| Hazardous ingredients<br>CAS-No.               | LogKow | Bioconcentration factor (BCF) | Exposure duration | Species                                     | Temperature        | Method                                                                                                                                        |
|------------------------------------------------|--------|-------------------------------|-------------------|---------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Phenol<br>108-95-2                             | 1.47   | 17,5                          | 5 h               | Danio rerio (reported as Brachydanio rerio) | 25 °C<br><br>30 °C | OECD Guideline 305 E (Bioaccumulation: Flow-through Fish Test)<br>OECD Guideline 117 (Partition Coefficient (n-octanol / water), HPLC Method) |
| 4-(1,1,3,3-tetramethylbutyl)phenol<br>140-66-9 | 4.5    |                               |                   |                                             | 23 °C              | OECD Guideline 107 (Partition Coefficient (n-octanol / water), Shake Flask Method)                                                            |

## 12.4 Results of PBT and vPvB assessment

| Hazardous ingredients<br>CAS-No.                    | PBT/vPvB                                                       |
|-----------------------------------------------------|----------------------------------------------------------------|
| Silver >= 99,9 % Ag in powder (< 1 mm)<br>7440-22-4 | Not fulfilling PBT (persistent/bioaccumulative/toxic) criteria |

## 12.5 Other adverse effects

No data available.

## SECTION 13: Disposal considerations

### 13.1 Waste treatment methods

#### Treatment of contaminated packaging

Disposal of the product:

Must be subjected to special treatment in consultation with the responsible local authority.

#### Waste code

080312

The waste codes of the European waste catalogue are based on the origin of the waste, not on the product. Thus, no waste code can be provided for products that are applied in different industries.

The waste codes supplied are to be regarded as recommendations for the user.

# Safety Data Sheet

according to Regulation (EC) No 1907/2006



Trade name: LPKF ProConduct Paste  
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## SECTION 14: Transport information

### 14.1 UN number

3082

### 14.2 UN proper shipping name

#### ADR/RID

ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Silver)

#### IMDG-Code / ICAO-TI / IATA-DGR

ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Silver)

### 14.3 Transport hazard class(es)

9

### 14.4 Packing group

III

### 14.5 Environmental Hazards

not applicable

#### Environmentally hazardous substances

ADR/RID / IMDG code / ICAO-TI / IATA-DGR: ☐ yes / ☒ no

Marine Pollutant: ☒ yes / ☐ no

### 14.6 Special precautions for user

Not applicable

Tunnel code: (E)

### 14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable

## SECTION 15: Regulatory information

### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

#### EU legislation

##### Directive 1999/13/EC (VOC Solvents Directive):

VOC content: 21.6 %

#### National regulations/information (Germany)

##### Water hazard classification

WGK = 3, highly water endangering product. Classification according to the mixture rules in German VwVwS regulation annex 4 from 27 July 2005.

##### Storage class according to TRGS 510

6.1C

##### General notes (DE):

This product is in scope of the German regulation "ChemikalienVerbotsVerordnung" (chemicals restriction act).

### 15.2 Chemical safety assessment

A chemical safety assessment has not been carried out.

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**SECTION 16: Other information**

**Full text of R-phases, hazard statements, safety advice and/or S-phrases referenced in chapters 2 to 15**

**According to 67/548/EEC:**

R10 Flammable.  
R23/24/25 Toxic by inhalation, in contact with skin and if swallowed.  
R34 Causes burns.  
R36 Irritating to eyes.  
R37 Irritating to respiratory system.  
R38 Irritating to skin.  
R41 Risk of serious damage to eyes.  
R48/20/21/22 Harmful: danger of serious damage to health by prolonged exposure through inhalation, in contact with skin and if swallowed.  
R50/53 Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.  
R51/53 Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.  
R65 Harmful: may cause lung damage if swallowed.  
R66 Repeated exposure may cause skin dryness or cracking.  
R67 Vapours may cause drowsiness and dizziness.  
R68 Possible risk of irreversible effects.

**According to Regulation (EC) No. 1272/2008:**

H301 Toxic if swallowed.  
H311 Toxic in contact with skin.  
H314 Causes severe skin burns and eye damage.  
H315 Causes skin irritation.  
H318 Causes serious eye damage.  
H319 Causes serious eye irritation.  
H331 Toxic if inhaled.  
H341 Suspected of causing genetic defects.  
H373 May cause damage to organs through prolonged or repeated exposure.  
H400 Very toxic to aquatic organisms.  
H410 Very toxic to aquatic life with long lasting effects.

**Further information**

The information contained in this safety data sheet is based on our present knowledge. However, this does not constitute a guarantee for any specific product features and does not establish a legally valid contractual relationship.

**Indication of changes since the last version**

Complete revision

**Department issuing this data sheet**

Product management